

EduPlatform Enterprise LMS

Final Architecture Guide v5

ASP.NET Core 8 · Clean Architecture · EF Core 8 · Docker · SQL Server · Redis · Serilog · Seq · xUnit

Final Portfolio Release

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Version: Final Portfolio Release

Stack: ASP.NET Core 8, EF Core 8, SQL Server 2022, Docker, Redis, Serilog, Seq, xUnit, Swagger, MVC, JWT

1. Executive Summary

EduPlatform is an enterprise-style Learning Management System that supports students, teachers, parents, administrators, and content managers. It contains two presentation surfaces:

1. EduPlatform.API - JWT-secured REST API for machine/mobile/front-end clients.
2. EduPlatform.Web - ASP.NET Core MVC portal using cookie authentication and Razor views.

Both surfaces share the same Infrastructure layer, database, Identity system, repositories, UnitOfWork, services, and seeders.

2. Solution Structure

```
EduPlatform/  
├─ EduPlatform.Domain  
├─ EduPlatform.Application  
├─ EduPlatform.Infrastructure  
├─ EduPlatform.API  
├─ EduPlatform.Web  
├─ EduPlatform.Tests  
├─ docs  
├─ scripts  
├─ docker-compose.yml  
└─ README.md
```

Dependency Rule

Domain <- Application <- Infrastructure <- API/Web

The Domain layer does not depend on EF Core, MVC, Identity, Docker, or any external framework.

3. Domain Layer

Contains:

- Entities
- Enums
- Role constants
- Repository interfaces
- UnitOfWork contract
- BaseEntity

Key entities include:

- Course
- Grade

- Subject
- Category
- Section
- Lesson
- Enrollment
- CartItem
- WishlistItem
- Order
- Payment
- Coupon
- Certificate
- LessonProgress
- LiveSession
- Notification
- CourseQuestion
- CourseAnswer
- ParentStudentLink

BaseEntity

Provides:

- `Id`
- `CreatedAt`
- `UpdatedAt`
- `CreatedBy`
- `UpdatedBy`
- `IsDeleted`

Soft-delete support protects academic/payment data from physical deletion.

4. Application Layer

Contains:

- DTOs
- Commands
- Validators
- Result wrappers
- Interfaces
- Mapping profile
- Pagination models

Key Patterns

- `Result`
- `ServiceResult<T>`
- `ApiResponse<T>`
- `PagedResult<T>`

- FluentValidation validators
- AutoMapper profile

Why DTOs?

DTOs prevent domain entities from leaking into API responses or MVC views.

5. Infrastructure Layer

Contains:

- `AppDbContext`
- Identity user
- Entity configurations
- Repositories
- UnitOfWork implementation
- LMS services
- Auth services
- Token service
- Email service
- Seeders

ApplicationUser

Extends IdentityUser with:

- FullName
- IsDeleted
- CreatedAt
- RefreshToken
- RefreshTokenExpiresAt

UnitOfWork

Provides one shared DbContext and one atomic commit for multi-step operations.

Seeders

Seeders create:

- roles
- admin account
- teachers
- students
- parents
- grades
- subjects
- categories
- courses
- sections/lessons
- enrollments
- demo LMS data

- parent/student links

6. API Layer

Responsibilities:

- REST API endpoints
- JWT authentication
- Swagger/OpenAPI
- Rate limiting
- CORS
- Validation filter
- Exception middleware
- Health checks

Middleware Order

1. ExceptionMiddleware
2. Swagger
3. Static files
4. Serilog request logging
5. CORS
6. Rate limiter
7. Authentication
8. Authorization
9. Health checks and controllers

Auth Endpoints

- `POST /api/Auth/register`
- `POST /api/Auth/login`
- `POST /api/Auth/refresh`
- `POST /api/Auth/revoke`
- `POST /api/Auth/forgot-password`
- `POST /api/Auth/reset-password`
- `GET /api/Auth/me`

7. Web Layer

Responsibilities:

- MVC portal
- Razor views

- ViewModels
- Cookie authentication
- role-based dashboard
- bilingual UI
- profile/change-password UI
- admin user access control

Key MVC Routes

Route	Purpose
`/Account/Login`	Login
`/Account/Profile`	User profile
`/Account/ChangePassword`	User changes own password
`/Account/Users`	Admin/ContentManager user access list
`/Account/CreateUser`	Admin creates users
`/Account/EditRoles/{id}`	Admin edits roles
`/Account/ResetPassword/{id}`	Admin resets password
`/PendingTasks`	Pending teachers/courses/locked users
`/Dashboard`	Role-based dashboard

8. Authentication Design

API

Uses JWT Bearer tokens.

Access token:

- contains user id
- email
- roles
- JTI
- 24-hour expiry by default

Refresh token:

- generated using cryptographic random bytes
- stored on the user account
- expires based on `Jwt:RefreshTokenDays`
- can be revoked

MVC

Uses ASP.NET Core Identity cookie auth.

Cookie:

- HttpOnly
- SameSite=Lax
- 8-hour sliding expiration

9. Authorization Model

Roles:

- Admin
- ContentManager
- Teacher
- PendingTeacher
- Student
- Parent

Admin

Can manage users, roles, passwords, unlock accounts, approve teachers/courses, and access all dashboards.

ContentManager

Can review users/pending tasks and unlock accounts, but does not have full user creation/reset power unless explicitly granted.

Teacher

Can create/manage own courses and live sessions.

Student

Can browse, enroll, track progress, review courses, and earn certificates.

Parent

Can view linked student learning status.

10. Database Strategy

Local Visual Studio

Use:

Server=localhost,1433

Docker Containers

Use:

Server=sqlserver,1433

Why?

- Local apps connect through the host port.
- Containers connect through Docker internal DNS service names.

11. Docker Architecture

Services:

- sqlserver

- redis
- seq
- api
- web

Ports:

- API: 8080
- Web: 8081
- SQL: 1433
- Seq: 8088
- Redis: 6379

12. Logging Strategy

Serilog writes to:

- Console
- Rolling files
- Seq

Seq allows searching logs by:

- request path
- status code
- trace id
- exception type
- application name

13. Validation Strategy

FluentValidation validates request DTOs/commands before controller execution.

ValidateModelFilter standardizes validation failures into a consistent API response.

14. Testing Strategy

Current unit tests cover:

- BaseEntity
- enums
- result wrappers
- DTOs
- ViewModels

Recommended next test layer:

- controller tests
- service tests
- repository tests
- integration tests with Testcontainers SQL Server

15. OBS / Live Video Preparation

OBS Studio setup was prepared for future LMS content creation:

- Display Capture
- Webcam
- Microphone
- 1080p / 30 FPS
- hardware encoding
- teaching scene

Future integration options:

- YouTube Live links
- Zoom/Teams links
- WebRTC
- Agora
- Jitsi
- recording upload to course lessons

16. Security Checklist

Before production:

- Move JWT key to environment variable or secret manager.
- Replace development SA password.
- Enable confirmed email.
- Add real email provider.
- Use HTTPS only.
- Add refresh-token rotation policy.
- Add audit logs for role/password changes.
- Add file upload validation.
- Add rate limits per endpoint category.

17. Final Evaluation

EduPlatform is now a strong enterprise-style project suitable for:

- GitHub portfolio
- technical interviews
- backend architecture demonstration

- full-stack .NET practice
- MVP demonstration

Overall status: strong portfolio / enterprise junior-to-mid level project.